

Seedless Condensers for Efficiently Samplable Sources

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Abstract

Is it possible to efficiently convert any arbitrary source with sufficiently high (min-)entropy into nearly uniformly random bits? Information-theoretically, this is achievable using seeded extractors, provided the source is independent of the seed. However, in the absence of any such independence guarantee, no solution is possible for general computationally unbounded sources. Even for efficiently samplable sources, we cannot *extract* randomness that is statistically close to uniform, but can at best *condense* the randomness into an output that is only missing logarithmically many bits of entropy compared to the uniform distribution. Dodis, Ristenpart and Vadhan (TCC '12) constructed such seeded condensers for efficiently samplable sources that can depend arbitrarily on the seed assuming near-optimal security of collision-resistant hash functions. In this work, we investigate whether such condensers can be made *seedless*. We present several new results:

- We construct seedless condensers for all efficiently samplable sources assuming near-optimal security of *keyless multi-collision resistant hash functions*. We justify this assumption by showing it holds in the auxiliary-input random oracle model.
- We show that such seedless condensers cannot be proven secure via a black-box reduction from any standard *game-based assumption*, even if we assume optimal exact security. In fact, we give a general black-box separation that applies to a broad class of seedless primitives, with seedless condensers as a special case.
- We consider the class of efficiently samplable *distributed* sources where two parties jointly sample randomness $x = (x_0, x_1)$, with one party honestly choosing x_b uniformly at random while the other party adversarially chooses x_{1-b} depending on x_b . We show how to construct seedless condensers for such sources assuming near-optimal security of game-based assumptions – either: (1) adaptive one-way functions, or (2) a certain form of (single-input) correlation-intractable hash functions that can be instantiated from key-dependent-message (KDM) secure encryption.

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1 Introduction

Is it possible to efficiently convert any arbitrary source of sufficiently high (min-)entropy into nearly uniformly random bits? Information-theoretically, this is achievable using *seeded randomness extractors* [NZ96], provided the source is independent of a uniformly random seed. In that case, one can extract almost all of the entropy in the source, and the extracted randomness is guaranteed to be statistically close to uniform even given the seed.

Unfortunately, if we remove the guarantee that the source is independent of the seed, or remove the seed entirely, then no non-trivial randomness extraction is possible: for any (seeded) extractor that acts on n -bit sources, there is a (seed-dependent) source with very high min-entropy $n - v$ for which the first v output bits of the extractor are completely fixed. Namely, the source is uniformly random over the 2^{n-v} preimages of some fixed v output bits. However, such bad sources may not be efficiently samplable in general. Can we get better guarantees if we restrict ourselves to efficiently samplable sources?

Unfortunately, it is still impossible to extract even one nearly uniformly random bit in this setting. In fact, for any logarithmic $v = O(\log n)$ there is a (seed-dependent) source that is efficiently samplable in time $\text{poly}(n)$ with min-entropy $n - v$ for which the first v extracted bits are completely fixed. Namely, the source repeatedly samples uniformly random values and uses rejection-sampling to output the first one that maps to some fixed v output bits. However, this strategy only allows us to fix logarithmically many bits and therefore leaves the possibility of very good (seeded) *randomness condensers* that take any efficiently-samplable (seed-dependent) source of sufficiently high min-entropy k , and concentrate the entropy into a short output of size $m \approx k$ that has nearly-maximal min-entropy $k' = m - O(\log n)$.

Note that such condensers would be extremely useful despite the fact that their output is not statistically close to uniform. As shown by [DY13], the output of such a condenser can be used in place of a uniformly random key in many cryptographic applications without sacrificing security. Namely, this holds for all *unpredictability applications*, where the goal of the attacker is to win some game with non-negligible probability, which includes (e.g.,) one-way functions, message-authentication, signatures, and sound arguments. It also holds for *square-friendly indistinguishability applications*, which include (e.g.,) symmetric-key or public-key CPA-secure encryption and weak PRFs. On the other hand, it does not include some other indistinguishability applications such as one-time symmetric-key encryption or PRGs. See [DY13] for a comprehensive discussion.

Prior work on seeded condensers. Dodis, Ristenpart, and Vadhan [DRV12] defined the notion of computational seeded condensers for efficient seed-dependent sources. They considered a scenario where a seed S is chosen uniformly at random and a seed-dependent source $X \leftarrow A(S)$ is chosen by a polynomial-size (adversarial) sampler A . They defined a seeded function $\text{Cond}(X; S)$ that outputs m bits to be a $(k \rightarrow_\varepsilon k')$ -condenser if, whenever X has k -bits of min-entropy given S then $\text{Cond}(X; S)$ is ε -statistically close to having k' bits of min-entropy given S . They show that any collision-resistant hash with m -bit output and near-optimal security, where polynomial-size attackers have at most a $\text{poly}(n)/2^m$ chance of finding a collision, gives such a condenser for $k \geq m - O(\log n)$ and $k' = m - O(\log n) - \log(1/\varepsilon)$. In particular, the output is close to only missing $O(\log n)$ bits of entropy, which is the best we can hope for, albeit this is only shown with statistical distance $1/\text{poly}(n)$.¹

¹The results of [DRV12] are more general and also apply to other notions of entropy and to less-optimal security levels of collision-resistance with corresponding parameters; see the paper for details.

1.1 Our Results

In this work, we ask whether we can get computational condensers in the *seedless* setting. Specifically, we are interested in functions $\text{Cond}: \{0, 1\}^n \rightarrow \{0, 1\}^m$ such that on input distributions X with min-entropy at least k , it holds that $\text{Cond}(X)$ has min-entropy at least k' . We call such functions $(k \rightarrow k')$ -*seedless condensers*. Seedless condensers are preferable over seeded condensers (for seed-dependent sources) for several reasons. Firstly, they remove the reliance on a trusted party to choose a random seed. Secondly, they remove the reliance on needing any source of uniform randomness altogether.

Seedless condensers from multi-collision resistance. We first show that any function that satisfies a strong form of *keyless* multi-collision resistance is a good seedless condenser. Bitansky, Kalai, and Paneth [BKP18] first formalized keyless ℓ -multi-collision resistant hashing (MCRH), which requires that for a fixed hash function $h: \{0, 1\}^n \rightarrow \{0, 1\}^m$, it is hard to efficiently find ℓ distinct inputs $x_1, \dots, x_\ell$ such that $h(x_1) = \dots = h(x_\ell)$. Unfortunately, this notion is impossible to achieve in the seedless setting for any polynomial ℓ ; an $(\ell \cdot n)$ -size non-uniform attacker could simply hardcode such an ℓ -wise collision.

To get around this, [BKP18] consider the above notion for values of ℓ which are sufficiently larger than the attacker's size. In this work, we consider a variant of this, which was also suggested by [BKP18], that requires the following:

For every polynomial-size attacker A , there is a polynomial-size set of inputs S such that A cannot find ℓ -wise collisions with all inputs *outside of* S .

We say that an ℓ -MCRH as above has security μ if, for any polynomial-size attacker, the success probability in the above game is bounded by at most μ .

Our first result shows that hash functions with strong ℓ -MCRH security are good condensers.

Theorem 1.1 (Informal; see Theorem 4.2). *Any keyless ℓ -MCRH with security $\mu = \delta^{\ell-1}$ is a $(k \rightarrow k')$ -seedless condenser for $k' = \min\{\log(1/\delta) \cdot (1 - 1/\ell), k - O(\log n)\}$.*

Our actual result is more general and also allows weaker notions of MCRH where the set S can be super-polynomial.

An optimally secure ℓ -MCRH has security $\mu = \delta^{\ell-1}$ for $\delta = \text{poly}(n) \cdot 2^{-m}$. Furthermore, we justify the existence of optimally secure ℓ -MCRH, by showing they exist in the auxiliary-input random oracle model (AI-ROM) of Unruh [Unr07], which captures non-uniform attackers in the ROM. This gives the following result for condensers.

Theorem 1.2 (Informal; see Theorem 4.5 and Corollary 4.7). *In the AI-ROM, there exist $(k \rightarrow k')$ -seedless condensers for $k' = \min\{m, k\} - O(\log n)$.*

In other words, if the input min-entropy $k \geq m$, the condensed output has almost full min-entropy $k' = m - O(\log n)$ and is only missing $O(\log n)$ bits of entropy compared to uniform. Otherwise, if we already start with a source having low entropy $k < m$, then the output has entropy $k' = k - O(\log n)$ and is only missing $O(\log n)$ bits of entropy compared to the source.

We emphasize two main advantages of our result compared to [DRV12]. First and most importantly, our condensers are seedless and don't require any external source of high-quality randomness.

Additionally, the output for our condensers truly has k' -bits of min-entropy, whereas the outputs for the condensers of [DRV12] are only $1/\text{poly}(n)$ -statistically close to having k' bits of entropy.²

We believe that building seedless condensers from multi-collision resistance or in the AI-ROM are important first steps towards basing condensers on plausible candidates, but this raises the following question:

Can we build seedless condensers from any standard cryptographic assumptions in the plain model?

Unfortunately, we give a negative answer to this question in the general case.

Black-box separations for seedless primitives. The security assumption on keyless MCRH we need for Theorem 1.1 is quite strong and non-standard. It does not even have the usual format of a “game-based assumption” in cryptography that is formulated as an interactive game between an attacker and a (potentially inefficient) challenger. Such game-based assumptions generalize “falsifiable” assumptions [Nao03, GW11], where the challenger is also required to be efficient. Instead, the challenger in the MCRH security definition depends on the attacker in a non-black-box way—this is captured by the extra quantifier where for every attacker there exists some attacker-dependent set S . Similarly, the condenser security definition does not have the format of a standard cryptographic game—the challenger has to verify the min-entropy of the adversarially-generated source distribution.

For our next result, we give a barrier for constructing a general class of seedless primitives like condensers or MCRH, where the adversary does not get any input and simply generates some (randomized) output. Specifically, we give a black-box separation of such primitives from all game-based assumptions as follows.

Theorem 1.3 (Informal; see Theorem 5.4). *For a broad class of seedless primitives, there is no black-box reduction of their security to any game-based assumption.*

We emphasize that the result above holds even if we assume arbitrarily strong levels of concrete security for the underlying assumptions.

A previous work of [Wic13] showed that seeded condensers for seed-dependent sources (with some restrictions) cannot be proven secure via a black-box reduction from a polynomially-secure game-based assumption. However, this does not rule out reductions that rely on strong concrete security of game-based assumptions, which is why [DRV12] was able to construct such condensers based on the strong concrete security of collision-resistant hash functions. Our negative result of Theorem 1.3 shows that seedless condensers must rely on fundamentally different assumptions altogether—not just stronger levels of security for standard assumptions.

Seedless condensers for distributed sources. With the above barrier in mind, we consider a restricted yet natural class of efficiently samplable sources that we call *distributed* sources. We think of two parties that jointly sample some randomness $x = (x_0, x_1) \in \{0, 1\}^{2k}$ where one party samples $x_b \in \{0, 1\}^k$ uniformly at random while the other party adversarially chooses $x_{1-b} \in \{0, 1\}^k$ depending on x_b . The difficulty of condensing such sources comes from the fact that the same function Cond must work for both choices of $b \in \{0, 1\}$. However, seedless condensers for such

²We note that if we start with *keyed* ℓ -MCRH with the same level of security, we get analogous results in the seed-dependent setting of [DRV12] where the output has truly k' bits of entropy. They considered this only for the case of standard collision-resistance where $\ell = 2$, which loses a factor of $(1 - 1/\ell) = 1/2$ in the output entropy.

distributed sources are not covered by our black-box separation result above and, indeed, we show how to construct them from strong game-based assumptions.

We give two constructions of seedless condensers for distributed sources. Our first construction is based on a pair of adaptive one-way functions, as introduced by [PPV08]. These are functions f_0, f_1 where f_b is one-way even given an inversion oracle of f_{1-b} . We also require that the functions satisfy an additional statistical property; roughly that every output corresponds to a similar number of inputs up to some small multiplicative factor r . We call such functions r -bounded regular.

Theorem 1.4 (Informal; see Theorem 6.4). *Assuming the existence of a pair of r -bounded-regular adaptive one-way functions with security μ , then there exists a $(k \rightarrow k')$ -seedless condenser for distributed sources for $k' = \log(1/\mu) - O(\log r)$.*

In particular, optimally-secure $\text{poly}(n)$ -bounded-regular adaptive one-way functions have $\mu = \text{poly}(n) \cdot 2^{-m}$ so $k' = m - O(\log n)$.

We consider potential candidates for f_0, f_1 based on hardness in different axes as suggested by [LPS17, KK19]. For instance, adaptive one-wayness holds if f_0 is quantumly broken in polynomial-time but classically exponentially-secure (e.g., based on the discrete-logarithm problem over elliptic curves), and f_1 is secure against quantum polynomial-time circuits but classically broken in sub-exponential-time (e.g., parallel repetition of a post-quantum cryptographic one-way hash).

Our second construction of a seedless condenser for distributed sources is based on a certain form of (seeded, single-input) *correlation-intractable* (CI) hash functions introduced by [CGH98]. A family of keyed hash functions h is CI for a relation R if, for a random hash key hk , it is hard to find an input x such that $(x, h(\text{hk}, x)) \in R$. Our construction relies on a particular CI hash function for all constant relations based on one-way permutations, as well as a CI hash function for all sparse relations that are efficiently recognizable, which gives the following result.

Theorem 1.5. *Assuming the existence of one-way permutations and CI hash functions for sparse, efficiently recognizable relations, both with security μ , then there exists a $(k \rightarrow k')$ -seedless condenser for distributed sources for $k' = \log(1/\mu)$.*

The CI hash construction from [CCRR18] based on particular encryption schemes with key-dependent message (KDM) security gives a candidate instantiation with optimal security $\mu = \text{poly}(n) \cdot 2^{-m}$. Combined with any one-way permutation with near optimal-security (e.g., discrete-log based), this results in a candidate seedless condenser for distributed sources with $k' = m - O(\log n)$; see Section 7.3 for further discussions on potential candidates for this construction.

1.2 Technical Overview

In this section, we give a high level overview for the main techniques of our results regarding seedless condensers. Recall that our goal is to construct efficient functions $\text{Cond}: \{0, 1\}^n \rightarrow \{0, 1\}^m$ for $m < n$ such that on input efficiently samplable distributions X with min-entropy k , it holds that $\text{Cond}(X)$ has min-entropy k' . We call such functions $(k \rightarrow k')$ -seedless condensers. We refer to Section 3 for our formal definitions for seedless condensers.

1.2.1 Condensers from Multi-Collision Resistance

We first show if Cond satisfies a notion of *keyless* multi-collision resistance, then it is a seedless condensers. Before formalizing the notion of collision-resistance we consider, let's see why this should intuitively be true.

Suppose that Cond is *not* a $(k \rightarrow k')$ -seedless condenser. This means that there is an efficiently samplable distribution X such that $\text{Cond}(X)$ has min-entropy at most k' . So, in particular, there must exist a value $z \in \{0, 1\}^m$ that $\text{Cond}(X)$ is likely to hit. Formally, it holds that

$$\Pr[\text{Cond}(X) = z] \geq 2^{-k'}.$$

This suggests a simple algorithm for finding collisions: sample many values $x_1, \dots, x_\ell$ from X and hope that $z = \text{Cond}(x_1) = \dots = \text{Cond}(x_\ell)$. If the min-entropy k of X is larger than k' , it follows that each successive value x_i will be distinct from $x_1, \dots, x_{i-1}$ and hit $z = \text{Cond}(x_i)$ with probability $\approx 2^{-k'}$. As a result, this simple algorithm will find an ℓ -wise collision with probability $\approx 2^{-k' \cdot \ell}$.

Now suppose that it were hard to find ℓ -wise collisions in Cond . In particular, for a random function, any efficient (uniform) attacker will succeed with probability at most $\text{poly}(n) \cdot 2^{-m \cdot (\ell-1)}$. Given our attack above, this would imply that

$$2^{-k' \cdot \ell} \leq \text{poly}(n) \cdot 2^{-m \cdot (\ell-1)} \Rightarrow k' \geq m \cdot (1 - 1/\ell) - O(\log n).$$

In other words, by setting $\ell = m$, this shows that the output of a random function Cond has nearly-maximal min-entropy—it's only missing $O(\log n)$ bits.

Multi-collision resistance in the seedless setting. Unfortunately, the above reasoning has a *key* flaw—multi-collision resistant hash functions don't exist in the seedless setting. Namely, there exists a non-uniform attacker that finds an ℓ -wise collision by simply hardcoding the collision in its non-uniform advice. To deal with this issue, we turn to a notion of *keyless* multi-collision resistance introduced by [BKP18] and require for every attacker A , there is a set of inputs S of size $\approx |A|$ such that A cannot find ℓ -wise collisions *outside* of S . It remains to argue that this notion is both (1) useful and (2) may feasibly exist.

For (1), we observe that this security notion does not significantly affect the reduction above. As long as k is larger than k' , it is also unlikely that each sampled value x_i will be in the set S of size $\text{poly}(n)$, so the attack will still succeed with probability $\approx 2^{-k' \cdot \ell}$. We refer to Section 4.1 for the formal proof of this claim.

For (2), we justify this assumption by proving that such functions exist in the auxiliary-input random oracle model (AI-ROM) of [Unr07]. In the AI-ROM, the attacker is given arbitrary—yet bounded size—non-uniform advice about a random function, so it more closely captures the standard notion of non-uniform security considered in Cryptography. To show that a random function is still ℓ -wise collision resistant in this model, we make use of the intermediate bit-fixing random oracle model also introduced by [Unr07], where some (attacker-dependent) points of the random function are fixed and the rest of the function can be lazily sampled. The set of fixed points can essentially be used as the set S in our definition. Then, our result follows from a lemma of Coretti, Dodis, Guo, and Steinberger [CDGS18] that relates the advantage of any attacker in the AI-ROM to the bit-fixing random oracle model; see Section 4.2 for full details.

Combining (1) and (2) above immediately gives our result for seedless condensers in the AI-ROM with near-maximal min-entropy $k' = m - O(\log n)$.

1.2.2 Black-Box Separations for Seedless Primitives

We next give evidence for why strong and non-standard assumptions like MCRH are necessary for building seedless condensers. Specifically, we show for a broad class of seedless primitives that there

is no black-box security reduction to game-based assumption. For simplicity, we focus on the case of seedless condensers in this overview; see Section 5 for the full proof and the formal notion for the seedless primitives we capture.

Suppose we could construct a $(k \rightarrow k')$ -seedless condenser $\text{Cond}: \{0, 1\}^n \rightarrow \{0, 1\}^m$ from any cryptographic primitive with security game $\mathcal{G}$. Recall that a security game $\mathcal{G}$ is defined by a (possibly inefficient) challenger that interacts with an attacker A and outputs a bit b to indicate whether or not the security notion was broken. A fully black-box reduction from the security of Cond to $\mathcal{G}$ is specified by an oracle circuit $B^{(\cdot)}$ such that the following holds: Given any (possibly inefficient) attacker A that breaks Cond , it holds that B^A breaks the security of $\mathcal{G}$.

Our main observation is that if such a reduction exists, then $\mathcal{G}$ is not secure against non-uniform attackers (which is the standard notion of security in Cryptography). Specifically, we show that the reduction B itself hardcoded with additional non-uniform advice x of size at most $|B|$ can break $\mathcal{G}$.

To see why this holds, consider an inefficient attacker A^* that breaks the security of Cond (such an attacker exists for non-trivial condensers, see Section 3.1). By definition, B^{A^*} breaks the security of $\mathcal{G}$. However, A^* is a stateless, randomized algorithm that doesn't take any inputs, so we can think of each of its outputs as independent samples from some fixed distribution. It follows by an averaging argument that there exists some fixed set of answers that cause B to break the game $\mathcal{G}$, so we can hardcode these answers into a new non-uniform circuit B' that also breaks $\mathcal{G}$ without even calling A^* .

1.2.3 Defining Distributed Sources

With the above barrier in mind, we turn to condensing a restricted yet natural class of efficiently samplable sources with min-entropy k . Specifically, we consider a scenario where two parties labeled 0 and 1 want to agree on a shared random string of some fixed length m . Each party provides a random string $x_0, x_1 \in \{0, 1\}^k$ and uses a condenser $\text{Cond}(x_0, x_1)$ to obtain the output. We want to ensure that the output still has high min-entropy even if one of the parties is corrupted and chooses their input x_b adversarially as a function of x_{1-b} .

Formally, we refer to such an input distribution (x_0, x_1) over $\{0, 1\}^n$ for $n = 2k$ as a *distributed source* which is specified by a stateful, efficient algorithm A and sampled via the following process:

- A outputs a bit $b \in \{0, 1\}$.
- The value $x_b \leftarrow \{0, 1\}^k$ is sampled honestly and uniformly.
- $A(x_b)$ computes a value $x_{1-b} \in \{0, 1\}^k$ arbitrarily as a function of x_b .

We refer to Section 3 for a formal definition of seedless condensers for distributed sources.

We note that constructing seedless condensers for such sources can be formulated as a security game, so it is not covered by the above black-box impossibility. Indeed, we next construct seedless condensers for such sources from primitives with game-based security definitions.

1.2.4 Condensers for Distributed Sources from Adaptive One-Wayness

We first show how to construct seedless condensers for distributed sources given a suitable pair of one-way functions f_0, f_1 . Specifically, we require that f_0 and f_1 are *adaptively* one-way as introduced by [PPV08], meaning that each function f_b remains one-way even given an oracle that inverts the other function f_{1-b} . For simplicity in this overview, we will assume that f_0, f_1 are additionally

permutations over $\{0,1\}^k$ so the output length $m = k$. In Section 6, we give the full proof and show how to relax this requirement to functions that are shrinking to $m < k$ bits and satisfy a relaxed “bounded regularity” property.

Given a pair of adaptive one-way functions f_0, f_1 , we construct a seedless condenser for distributed sources as follows:

$$\text{Cond}(x_0, x_1) := f_0(x_0) \oplus f_1(x_1).$$

Now suppose that Cond is not a $(k \rightarrow k')$ -seedless condenser. There must exist an algorithm A that samples a distributed source (x_0, x_1) and a value $z \in \{0,1\}^k$ such that

$$\Pr[\text{Cond}(x_0, x_1) = f_0(x_0) \oplus f_1(x_1) = z] \geq 2^{-k'}. \quad (1.1)$$

By symmetry and without loss of generality, it suffices to consider the case where A always specifies the bit $b = 0$, so x_0 is sampled honestly and uniformly. In this case, we will show how to use A to break the adaptive one-wayness of f_1 given an inverter for f_0 .

At a high level, we will plant the one-way permutation challenge y for f_1 such that $\text{Cond}(x_0, x_1) = f_0(x_0) \oplus y = z$, which means that $y = f_0(x_0) \oplus z$. However, A expects to receive the honestly sampled x_0 as input, so we will use the inverter for f_0 to compute $x_0 = f_0^{-1}(y \oplus z)$ and feed this to A —this is why we need to rely on the *adaptive* security of f_0, f_1 .

More formally, our reduction algorithm B is defined as follows: On input y , B computes $x_0^* = f_0^{-1}(y \oplus z)$ and outputs $x_1^* = A(x_0^*)$. Thus, the success probability of B is given by the following experiment:

$$\Pr \left[\begin{array}{l} x \leftarrow \{0,1\}^k \\ y = f_1(x) \\ x_0^* = f_0^{-1}(y \oplus z) \\ x_1^* = A(x_0^*) \end{array} : f_1(x_1^*) = y \right]$$

However, note that by construction, it holds that

$$f_1(x_1^*) = y \Leftrightarrow x_0^* = f_0^{-1}(f_1(x_1^*) \oplus z) \Leftrightarrow f_0(x_0^*) \oplus f_1(x_1^*) = z,$$

so the probability that B wins is $\Pr[\text{Cond}(x_0^*, x_1^*) = z]$. We claim, however, that this is exactly equal to the probability from Equation 1.1. This is because x_0^* is uniformly distributed since x is sampled uniformly and f_1 and f_0^{-1} are permutations. It follows that x_0 and x_0^* are identically distributed, so $x_1 = A(x_0)$ and $x_1^* = A(x_0^*)$ are identically distributed, which implies that

$$\Pr[\text{Cond}(x_0^*, x_1^*) = z] = \Pr[\text{Cond}(x_0, x_1) = z] \geq 2^{-k'}.$$

Thus, if f_0, f_1 cannot be inverted with probability better than μ in the adaptive one-way security game, then $2^{-k'} \leq \mu$ and $k' \geq \log(1/\mu)$. In particular, if f_0, f_1 have near-optimal security, so $\mu = \text{poly}(n) \cdot 2^{-k}$, it follows that $\text{Cond}(x_0, x_1)$ has near-maximal min-entropy $k' \geq k - O(\log n)$. We refer to Section 6.3 for more discussion on candidates for the pair of functions f_0, f_1 .

1.2.5 Condensers for Distributed Sources from Correlation Intractability

We next construct a seedless condenser Cond for distributed sources based on correlation-intractable (CI) hash functions. A hash function h is said to be CI for a relation R if on input a random hash

key hk , it is hard to find an input x with output $y = h(hk, x)$ such that $(x, y) \in R$. Notably, CI hash functions are *keyed* primitives, whereas our other constructions have all relied on un-keyed primitives.

Recall that a seedless condenser **Cond** for a distributed sources takes in two inputs (x_0, x_1) where we are guaranteed that one of x_0 or x_1 is uniformly random, but the hardness come from the fact that **Cond** doesn't *a priori* know which input will be random. Our main idea is compose two CI hash functions using x_0 and x_1 as the keys. So, let h_0, h_1 be keyed hash functions, then we define **Cond** as follows:

$$\text{Cond}(x_0, x_1) := h_0(x_0, h_1(x_1, x_0)).$$

Namely, we use x_0 as the hash key for the function h_0 and x_1 as the hash key for the function h_1 .

We again use the fact that if **Cond** is not a $(k \rightarrow k')$ -seedless condenser, there must exist an algorithm A that samples (x_0, x_1) and a value $z \in \{0, 1\}^m$ such that

$$\Pr[h_0(x_0, h_1(x_1, x_0)) = z] \geq 2^{-k'}.$$

We show that in the case where either x_0 or x_1 is random, this violates CI for h_0 or h_1 , respectively, with respect to some set of relations to be determined.

- First, suppose x_0 is random, which corresponds to the hash key for h_0 . Then x_1 implicitly specifies an input $x = h_1(x_1, x_0)$ such that $h_0(x_0, x) = z$. This corresponds to the constant relation $R_z = \{(x, y) : y = z\}$.
- Now, instead suppose x_1 is random, corresponding to the hash key for h_1 . Then the input $x = x_0$ results in an output $y = h_1(x_1, x)$ such that $h_0(x, y) = z$. This corresponds to the relation $R'_z = \{(x, y) : h_0(x, y) = z\}$.

So, it suffices to find candidates h_0, h_1 such that h_0 is simultaneously CI for all constant relations $\{R_z\}_z$ and h_1 is CI for all relations $\{R'_z\}_z$ as above. We refer to Section 7.2 for the full details of this security reduction.

For h_0 , we use a specific CI hash function based on any one-way permutation f . Namely, we define

$$h_0(hk, x) := f(x) \oplus hk.$$

Finding x such that $f(x) \oplus hk = z$ for any z and $hk \leftarrow \{0, 1\}^k$ is as hard as inverting f on a random $y = hk \oplus z \in \{0, 1\}^k$. Thus, h_0 is CI for all constant relations assuming f is a secure one-way permutation.

For h_1 , note that we can rewrite R'_z given our candidate for h_0 as follows:

$$R'_z = \{(x, y) : f(y) \oplus x = z\}.$$

As f is a permutation, this implies that for each relation specified by z and input x , there is at most one output y in the relation R'_z . This is known as a “sparse” relation, and the works of [CCR16, KRR17, CRR18] give candidate CI hash functions for such sparse relations. We note that the fact that R'_z is sparse depends on our specific choice for h_0 above, and even though constant relations are sparse, using the existing CI hash functions for all sparse relations for h_0 would not ensure that R'_z is sparse. This is why we require two different candidates for h_0 and h_1 .

Finally, we note that there is no loss in the security reductions above, so it follows that if the associated primitives and corresponding CI hash functions are broken with probability at most μ ,

then $k' \geq \log(1/\mu)$. If our candidates are secure for optimal values of $\mu = \text{poly}(n) \cdot 2^{-m}$, this implies that $\text{Cond}(x_0, x_1)$ will have near maximal min-entropy $k' \geq m - O(\log n)$. We refer to Section 7.3 for more discussion on candidates for the CI hash functions.

2 Preliminaries

We let $\mathbb{N} = \{1, 2, 3, \dots\}$ denote the set of natural numbers, and for any $n \in \mathbb{N}$, we write $[n]$ to denote the set $[n] = \{1, \dots, n\}$. For integers $a, b \in \mathbb{Z}$ with $a \leq b$, we write $[a, b]$ to denote the set $\{a, a+1, \dots, b\}$.

In this work, we primarily think of algorithms as being implemented by circuits, and we use size to refer to the run-time or number of gates in the circuit. For a circuit C , we write $|C|$ to denote the size of the circuit. For a function f , we write $|f|$ to denote the size of the circuit that implements f . We consider oracle algorithms defined with respect to some oracle $\mathcal{O}$ which are implemented by circuits that contain singular gates that implement the functionality of $\mathcal{O}$, so each call to the oracle only adds one to the circuits overall size.

For a distribution D , we write $x \leftarrow D$ to denote the process of sampling a value x from the distribution D . For a set S , we use $x \leftarrow S$ to denote the process of sampling a value x from the uniform distribution over S . We use $x = A(\cdot)$ to denote the output of a deterministic algorithm and $x \leftarrow A(\cdot)$ to denote the output of a randomized algorithm. We write $x := y$ to denote the assignment of value y to x .

For a distribution D , we define $\text{Supp}(D)$ to denote the support of the distribution D . We define $H_\infty(D)$ to denote the min-entropy of a distribution D , which is defined as

$$H_\infty(D) = -\log \left(\max_{x \in \text{Supp}(D)} \Pr[D = x] \right).$$

2.1 One-Way Functions and Permutations

A one-way function is an efficiently computable function that is hard to invert on a random output. If the function is additionally a permutation, we refer to it as a one-way permutation. We formalize these primitives as follows.

Definition 2.1 (One-Way Functions and Permutations). Let $n, m, t \in \mathbb{N}$ and $\mu \in [0, 1]$. We say that an efficiently computable function $f: \{0, 1\}^n \rightarrow \{0, 1\}^m$ is a (t, μ) -secure one-way function if for every t -size algorithm A , it holds that

$$\Pr \left[\begin{array}{l} x \leftarrow \{0, 1\}^n \\ y = f(x) \\ x' \leftarrow A(y) \end{array} : f(x') = y \right] < \mu.$$

If $n = m$ and f is additionally a permutation, we say that it is a (t, μ) -secure one-way permutation.

We emphasize that we do not consider *families* of one-way functions or permutations in this work, as our focus is primarily on seedless primitives. While families of one-way functions generically imply (seedless) one-way functions, this comes at a loss of security in the output length, which turns out to be important for the notion of seedless condensers we consider in this work.

3 Seedless Condensers

A seedless condenser is an efficient, deterministic function that compresses randomly generated strings. At a high level, we require if the input is computed efficiently and has some min-entropy, then the compressed output should also have some min-entropy guarantee. We define this notion first for general efficiently samplable min-entropy sources and then for a special type of such sources that naturally arise in distributed settings.

Seedless condensers for general sources. We first define min-entropy sources that are efficiently samplable, which captures the most general input for seedless condensers.

Definition 3.1 (Efficiently Samplable Min-Entropy Sources). Let $n, t, k \in \mathbb{N}$. A distribution X over $\{0, 1\}^n$ is a (t, k) -min-entropy source if $H_\infty(X) \geq k$ and there exists a t -size algorithm that perfectly samples from X .

We now define a seedless condenser for general min-entropy sources as follows.

Definition 3.2 (Seedless Condensers). Let $n, m, t, k, k' \in \mathbb{N}$. An efficient function $\text{Cond}: \{0, 1\}^n \rightarrow \{0, 1\}^m$ is a $(t, k \rightarrow k')$ -seedless condenser if for every (t, k) -min entropy source X over $\{0, 1\}^n$, it holds that $H_\infty(\text{Cond}(X)) \geq k'$.

Seedless condensers for distributed sources. We next define a distributed source, which is a special case of an efficiently samplable min-entropy source that naturally arises in a distributed setting. Specifically, consider a non-interactive 2-party protocol where each party publishes a k -bit string. Distributed sources capture a setting where one party is adversarial and chooses their string after seeing the honest party's string.

Definition 3.3 (Distributed Sources). Let $k, t \in \mathbb{N}$. We say that a distribution $X = (X_0, X_1)$ over $\{0, 1\}^{2k}$ is a (t, k) -distributed source if there exists a stateful t -size algorithm A such that X is identically distributed to the following process:

- Compute a bit $b \leftarrow A$.
- Sample $x_b \leftarrow \{0, 1\}^k$.
- Compute $x_{1-b} \leftarrow A(x_b)$ such that $x_{1-b} \in \{0, 1\}^k$.
- Output (x_0, x_1) .

Note that a (t, k) -distributed source is a (t, k) -min-entropy source over a domain of size $n = 2k$. This is because any string $(x_0, x_1) \in \{0, 1\}^{2k}$ can be output with probability at most 2^{-k} .

We define seedless condensers for distributed sources as follows.

Definition 3.4 (Seedless Condensers for Distributed Sources). Let $t, m, k, k' \in \mathbb{N}$. An efficient function $\text{Cond}: \{0, 1\}^{2k} \rightarrow \{0, 1\}^m$ is an $(t, k \rightarrow k')$ -seedless condenser for distributed sources if for every (t, k) -distributed source X it holds that $H_\infty(\text{Cond}(X)) \geq k'$.

3.1 Impossibility for Inefficiently Samplable Sources

We note that seedless condensers only make sense in the computational setting where the sources are efficiently samplable. Namely, we show for every function Cond , there always exists a large enough time bound t for which Cond is not a good condenser for sources samplable in time t . In this sense, any condenser implies some form of computational hardness.

By a “good” condenser, we mean that entropy deficit of the output $m - k'$ is smaller than the entropy deficit of the input $n - k$. In particular, we do not rule out “trivial condensers” where the input has almost full entropy ($n \approx k$) or there is almost no shrinkage ($n \approx m$).

We note that the proof below is closely related to similar impossibility results for (seeded) extractors (see, e.g., [CG88]).

Theorem 3.5. *For any $n, m, k, k' \in \mathbb{N}$ with $k \leq n$ and $k' \leq m$ such that $n - k \geq m - k' - 1$ and function $\text{Cond}: \{0, 1\}^n \rightarrow \{0, 1\}^m$, there exists a $t \in \mathbb{N}$ such that Cond is not a $(t, k \rightarrow k')$ -seedless condenser.*

Proof. By an averaging argument, there must exist a set of values $S \subseteq \{0, 1\}^m$ of size $|S| = 2^{k'-1}$ such that $|\text{Cond}^{-1}(S)| \geq 2^{n-m+k'-1}$. Define the distribution X to be a uniformly random element from $\text{Cond}^{-1}(S)$, which can be sampled in time $2^n \cdot |\text{Cond}|$ by running Cond on every input. Note that

$$H_\infty(X) = \log |\text{Cond}^{-1}(S)| \geq n - m + k' - 1 \geq k$$

by assumption. However, by construction, it also holds that

$$H_\infty(\text{Cond}(X)) \leq k' - 1 < k'$$

since $\text{Supp}(\text{Cond}(X)) = S$ and $|S| = 2^{k'-1}$. It follows that Cond is not a $(t, k \rightarrow k')$ -seedless condenser for $t = 2^n \cdot |\text{Cond}|$. $\square$

4 Seedless Condensers from Keyless Multi-Collision Resistance

In this section, we prove that a certain form of keyless multi-collision resistance implies seedless condensers for general min-entropy sources. Specifically, we require that t -size algorithms cannot find an ℓ -multi-collision that completely lies outside of a set of s inputs (that may arbitrarily depend on the adversary), which we formalize below.

Definition 4.1 (Keyless Multi-Collision Resistance). Let $t, s, \ell \in \mathbb{N}$ and $\mu \in [0, 1]$. We say that a function $h: \{0, 1\}^n \rightarrow \{0, 1\}^m$ is (t, s, ℓ, μ) -multi-collision resistant if for every t -size algorithm A , there exists a set S of size s such that

$$\Pr \left[\begin{array}{l} x_1, \dots, x_\ell \notin S \\ (x_1, \dots, x_\ell) \leftarrow A : \wedge x_1, \dots, x_\ell \text{ all distinct} \\ \wedge h(x_1) = \dots = h(x_\ell) \end{array} \right] < \mu.$$

The definition above is a minor modification of the notion from [BKP18] for $s \cdot n \geq t$. Their definition requires that A cannot output a multi-collision larger than its non-uniform advice. Intuitively in our definition, the set S plays the role of A ’s non-uniform advice, and we require that A cannot find too large of a multi-collision outside of this set.

We prove that Definition 4.1 implies a seedless condenser in Section 4.1, and in Section 4.2, we justify this multi-collision-resistance notion in the auxiliary-input random oracle model of Unruh [Unr07].

4.1 Multi-Collision-Resistant Hash Functions are Condensing

Our construction of a seedless condenser is vacuous—we show that any multi-collision-resistant hash function with strong enough security is already a condenser. Specifically, we require that finding an ℓ -multi-collision is $\mu = \delta^{\ell-1}$ hard for some parameter δ . Intuitively, this corresponds to a situation where the “best-possible” attack is to independently find $\ell - 1$ collisions with an initial output value, where the individual attack succeeds with probability at most δ . In this sense, the required assumption is qualitatively stronger for larger values of ℓ . As a function of ℓ , the min-entropy of the output grows with a factor of $(1 - 1/\ell)$, so we get better guarantees for larger values of ℓ . We formalize this implication in the following theorem.

Theorem 4.2. *Let $t, s, \ell \in \mathbb{N}$ and $\delta \in [0, 1]$. Any efficiently computable hash function $h: \{0, 1\}^n \rightarrow \{0, 1\}^m$ that is $(t \cdot \ell, s, \ell, \delta^{\ell-1})$ -multi-collision resistant is a $(t, k \rightarrow k')$ -seedless condenser for any k with $k' = \min\{\log(1/\delta) \cdot (1 - 1/\ell) - 1, k - \log(s + \ell) - 1\}$.*

Proof. Suppose h is not a $(t, k \rightarrow k')$ -seedless condenser. That implies that there exists a t -size algorithm A that samples a distribution X over $\{0, 1\}^n$ with $H_\infty(X) \geq k$ such that $H_\infty(h(X)) < k'$. In particular, there must exist a string $z \in \{0, 1\}^m$ such that

$$\Pr[x \leftarrow A : h(x) = z] \geq 2^{-k'}.$$

Given the algorithm A as above, we construct an attacker B against the ℓ -multi-collision resistance of h as follows:

Algorithm 4.3. B :

1. Sample $x_i \leftarrow A$ for each $i \in [\ell]$.
2. Outputs the tuple $(x_1, \dots, x_\ell)$.

Note that B clearly runs in time $t \cdot \ell$ since it simply runs A ℓ times. It remains to analyze the success probability of B in outputting a valid ℓ -multi-collision.

Let S be any set of size s . For each $i \in [\ell]$, define S_i to be the set $S \cup \{x_1, \dots, x_{i-1}\}$ defined after sampling the first $i - 1$ values from A . Let E_i be the event that $h(x_i) = z$ and x_i is not in S_i . It follows from the entropy bounds on X and $h(X)$ that

$$\begin{aligned} \Pr[E_i] &\geq \Pr[h(x_i) = z] - \Pr[x_i \in S_i] \\ &\geq 2^{-k'} - (s + \ell) \cdot 2^{-k} \\ &\geq 2^{-k'-1}, \end{aligned}$$

where the last line follows since $k' \leq k - \log(s + \ell) - 1$ by assumption. Whenever E_i holds for all $i \in [\ell]$, it follows that B finds a successful ℓ -multi-collision. Since these events are independent, it follows that B succeeds with probability at least

$$\prod_{i=1}^{\ell} \Pr[E_i] \geq (2^{-k'-1})^{\ell},$$

which is at most $\delta^{\ell-1}$ by assumption. Rearranging the implied inequality that $(2^{-k'-1})^\ell < \delta^{\ell-1}$, we get the following bound for k' as required:

$$k' > (1 - 1/\ell) \cdot \log(1/\delta) - 1.$$

□

4.2 Keyless Multi-Collision Resistance in AI-ROM

We next prove that there exists a keyless multi-collision resistant hash function satisfying Definition 4.1 in the *auxiliary-input* random oracle model (AI-ROM) of Unruh [Unr07]. In this model, the adversary is given arbitrary—yet bounded length—non-uniform advice about a random oracle. This is captured via a pre-processing phase where an unbounded algorithm Z takes as input a randomly sampled function and outputs bounded-size advice string for an adversary in a specified security game.

To help analyze the AI-ROM, [Unr07] introduced an intermediate *bit-fixing* random oracle model, where a partial assignment to the random oracle is fixed as a function of the adversary's advice string and the remainder of the function table can be sampled lazily and independently of the adversary. For a random function $R: \{0, 1\}^n \rightarrow \{0, 1\}^m$, we define a partial function P that specifies some number p of fixed input/output values. We then write $R[P]$ to denote a random function fixed at the points in P . We make use of the following lemma from Coretti, Dodis, Guo, and Steinberger [CDGS18] that relates the advantage of any adversary in the AI-ROM to the bit-fixing random oracle model.

Lemma 4.4 ([CDGS18]). *Let $R: \{0, 1\}^n \rightarrow \{0, 1\}^m$ be a uniformly random function and Z be any function with q -bit output. For any $\gamma > 0$, there exists a function F that outputs partial function assignments for p points such that for any oracle algorithm $A^{(\cdot)}$ that makes at most $t < p$ queries and outputs a single bit, it holds that*

$$\Pr [A^R(Z(R)) = 1] \leq 2^{(q+2 \log(1/\gamma)) \cdot t/p} \cdot \Pr [A^{R[F(Z(R))]}(Z(R)) = 1] + 2\gamma.$$

To use this lemma, we observe that the bit-fixing ROM essentially captures our adversarial game for multi-collision resistance in Definition 4.1. Namely, if we set the set of inputs S that the adversary must avoid in the multi-collision to be the set of fixed points $P = F(Z(R))$ in the bit-fixing ROM, then it suffices to analyze the probability of finding an ℓ -multi-collision in the plain random oracle model among the other lazily sampled outputs. Using this approach, we show in the following theorem that there exists a multi-collision resistant hash function in the AI-ROM.

Theorem 4.5. *Let $R: \{0, 1\}^n \rightarrow \{0, 1\}^m$ be a uniformly random function and Z be any function with q -bit output. For any $\ell \in \mathbb{N}$ and oracle algorithm $A^{(\cdot)}$ that makes at most t queries and outputs ℓ n -bit strings, there exists a set S of size $s = (q + 2 \cdot \ell \cdot m) \cdot t$ such that*

$$\Pr \left[(x_1, \dots, x_\ell) \leftarrow A^R(Z(R)) : \begin{array}{l} x_1, \dots, x_\ell \notin S \\ \wedge x_1, \dots, x_\ell \text{ all distinct} \\ \wedge R(x_1) = \dots = R(x_\ell) \end{array} \right] \leq 2 \cdot (t + \ell + 1) \cdot \left(\frac{t + \ell}{2^m} \right)^{\ell-1}.$$

Proof. Fix a value $\gamma = ((t + \ell)/2^m)^{\ell-1}$ and a corresponding function F that outputs a partial assignment for p points where $p = (q + 2 \cdot \ell \cdot m) \cdot t$ as guaranteed by Lemma 4.4. Given an oracle algorithm A from the theorem statement, we define an oracle algorithm $B^{(\cdot)}$ that takes as input a string $z \in \{0, 1\}^q$ and outputs a single bit as follows:

Algorithm 4.6. $B^R(z)$:

1. Sample inputs $(x_1, \dots, x_\ell) \leftarrow A^R(z)$.
2. Compute the set of points $P = F(z)$.
3. Output 1 if and only if $x_1, \dots, x_\ell \notin P$, $x_1, \dots, x_\ell$ are all distinct, and $R(x_1) = \dots = R(x_\ell)$.

Observe that $\log(1/\gamma) \leq \ell \cdot m$, so $(q + 2\log(1/\gamma)) \cdot t/p \leq 1$. Then, by Lemma 4.4 and setting $z = Z(R)$ and $P = F(z)$, it follows that

$$\Pr[B^R(z) = 1] \leq 2 \cdot \Pr[B^{R[P]}(z) = 1] + 2 \cdot \left(\frac{t + \ell}{2^m}\right)^{\ell-1}.$$

It is immediate by construction that $B^R(z) = 1$ if and only if the event from the theorem statement holds by letting S be the inputs for the partial assignment P of $s = p = (q + 2 \cdot \ell \cdot m) \cdot t$ many inputs. As a result, it suffices to analyze the probability that $B^{R[P]}(z) = 1$ in the bit-fixing ROM.

In the event that $B^{R[P]}(z) = 1$, recall that R can be lazily sampled at points outside of P , and in particular, $R(x_1), \dots, R(x_\ell)$ can all be sampled uniformly at random from $\{0, 1\}^m$. As B makes at most $t + \ell$ queries outside of P , the probability that B outputs 1 is at most the probability that there exists a set of ℓ colliding queries over the output space $\{0, 1\}^m$. Taking a union bound over each subset of ℓ queries, it follows that

$$\Pr[B^{R[P]}(z) = 1] \leq (t + \ell)^\ell \cdot 2^{-m \cdot (\ell-1)}.$$

It follows that

$$\Pr[B^R(z) = 1] \leq 2 \cdot (t + \ell + 1) \cdot \left(\frac{t + \ell}{2^m}\right)^{\ell-1},$$

as required for the theorem. □

4.3 Seedless Condensers in the AI-ROM

We get a seedless condenser in the AI-ROM as an immediate corollary to Theorems 4.2 and 4.5.

Corollary 4.7. *For any $n, m, t, k \in \mathbb{N}$ such that $k \leq n$, there exists a $(t, k \rightarrow k')$ -seedless condenser $\text{Cond}: \{0, 1\}^n \rightarrow \{0, 1\}^m$ in the AI-ROM for $k' = \min\{k, m\} - O(\log(t \cdot m))$.*

Proof. Let $R: \{0, 1\}^n \rightarrow \{0, 1\}^m$ be a random function modeled in the AI-ROM and we consider Cond to simply be the function R . We instantiate Theorem 4.5 with $\ell = m$ and consider oracle circuits with description size at most $t \cdot m$ that have at most $t \cdot m$ oracle gates. It follows that R is $(t \cdot m, s, m, \delta^{m-1})$ -multi-collision resistant for $s \leq t \cdot m \cdot (t \cdot m + 2m^2)$ and $\delta \leq 6t \cdot m/2^m$. By Theorem 4.2, it holds that R is a $(t, k \rightarrow k')$ -seedless condenser for any k with $k' = \min\{\log(1/\delta) \cdot (1 - 1/m) - 1, k - \log(s + m) - 1\}$. As $\log(1/\delta) \geq m - \log t - \log m - 3$ and $\log(s + m) \leq 2 \log t + 3 \log m + 2$, it follows that

$$k' \geq \min\{m - \log t - \log m - 5, k - 2 \log t - 3 \log m - 3\},$$

as required for the corollary. □

5 Black-Box Separation For Seedless Primitives

In this section we show a black-box separation for seedless condensers and keyless MCRH from any game-based cryptographic assumption. We define game based assumptions as follows:

Definition 5.1 (Cryptographic Game [HH09]). A cryptographic game $\mathcal{G} = (\Gamma, \gamma)$ is defined by a (possibly inefficient) randomized algorithm Γ , called the *challenger*, and a constant $\gamma \in [0, 1]$. The challenger Γ interacts with an algorithm A , the *attacker*, and outputs a bit b . We denote this interaction by $b = (A \leftrightarrow \Gamma)$. The *advantage* of an attacker A in the game $\mathcal{G}$ is defined as

$$\mathbf{Adv}_{\mathcal{G}}^A \stackrel{\text{def}}{=} \Pr[(A \leftrightarrow \Gamma) = 1] - \gamma.$$

A cryptographic game $\mathcal{G}$ is (t, μ) -secure if for all t -size attackers A , the advantage $\mathbf{Adv}_{\mathcal{G}}^A < \mu$.

Note that the seedless primitives we consider in this paper don't fall under the above definition. For the case of keyless MCRH, the challenger Γ depends on a set S where the set S depends on the attacker in a non-black-box way. For the case of seedless condensers, verifying that the source distribution has high min-entropy requires non-black-box access to the distribution.

In general, we observe that attackers against keyless MCRH or seedless condensers are randomized algorithms that take in no explicit inputs. We generalize this class of cryptographic primitives as follows:

Definition 5.2 (Seedless Cryptographic Property). A seedless cryptographic property $\mathcal{P}$ is a mapping that assigns to any no-input randomized algorithm A a bit $\mathcal{P}(A) \in \{0, 1\}$. A seedless cryptographic property $\mathcal{P}$ is t -secure if for all t -size *no-input* randomized circuits A , it holds that $\mathcal{P}(A) = 0$. We say that $\mathcal{P}$ is *breakable* if there exists such a circuit A^* of any size such that $\mathcal{P}(A^*) = 1$.

We note that keyless MCRH and seedless condensers fit into the above framework, with properties defined as follows:

- For a fixed $\ell, s \in \mathbb{N}$, $\mu \in [0, 1]$, and a keyless MCRH candidate $h: \{0, 1\}^n \rightarrow \{0, 1\}^m$. We define the mapping $\mathcal{P}^h$ (for the function h) as follows: For a no-input randomized algorithm A , we define $\mathcal{P}^h(A) = 0$ if there exists a set S of size s such that

$$\Pr \left[\begin{array}{l} (x_1, \dots, x_\ell) \leftarrow A : \\ \quad \wedge x_1, \dots, x_\ell \text{ all distinct} \\ \quad \wedge h(x_1) = \dots = h(x_\ell) \end{array} \right] < \mu.$$

Otherwise, we define $\mathcal{P}^h(A) = 1$ (meaning A contradicts the security of h).

- Similarly, for a fixed k and k' and a seedless condenser candidate $\text{Cond}: \{0, 1\}^n \rightarrow \{0, 1\}^m$, we define the mapping $\mathcal{P}^{\text{Cond}}$ as follows: For a no-input randomized algorithm A , we define $\mathcal{P}^{\text{Cond}}(A) = 1$ if for the distribution X output by A , it holds that $H_\infty(X) \geq k$ and $H_\infty(\text{Cond}(X)) < k'$. Otherwise, we define $\mathcal{P}^{\text{Cond}}(A) = 0$.

Note that $\mathcal{P}^h$ is t -secure iff h is (t, s, ℓ, μ) -keyless MCRH. And similarly, $\mathcal{P}^{\text{Cond}}$ is t -secure iff Cond is a $(t, k \rightarrow k')$ -seedless condensers.

We now define the notion of black-box reductions from any seedless primitives to any standard cryptographic game.

Definition 5.3 (Black-box reductions from seedless cryptographic property to a cryptographic game). Let $\mathcal{P}$ be a seedless cryptographic property and $\mathcal{G}$ be any cryptographic game. A t -size circuit with oracle access $B^{(\cdot)}$ is a (t, μ) -black-box reduction if for any (possibly inefficient) no-input randomized algorithm A such that $\mathcal{P}(A) = 1$, it holds that $\mathbf{Adv}_{\mathcal{G}}^{B^A} \geq \mu$.

We prove the following theorem, which implies our impossibility result.

Theorem 5.4. *Let $\mathcal{P}$ be a seedless cryptographic property that is breakable, and let $\mathcal{G}$ be any cryptographic game. If there exists a (t, μ) -black-box reduction from $\mathcal{P}$ to $\mathcal{G}$, then $\mathcal{G}$ is not $(2t, \mu)$ -secure.*

Proof. Since $\mathcal{P}$ is breakable, it holds that there exists a (possibly inefficient) circuit A^* for which it holds that $\mathcal{P}(A^*) = 1$. Let $B^{(\cdot)}$ be a (t, μ) -black-box reduction from $\mathcal{P}$ to $\mathcal{G} = (\Gamma, \gamma)$. By definition of $B^{(\cdot)}$ and $\mathcal{G}$, it holds that

$$\mathbf{Adv}_{\mathcal{G}}^{B^{A^*}} = \Pr \left[\left(B^{A^*} \leftrightarrow \Gamma \right) = 1 \right] - \gamma \geq \mu. \quad (5.1)$$

By the definition of seedless cryptographic property, the adversary A^* is a no-input randomized circuit. Assume first that on a call to the adversary, the adversary samples fresh randomness r and computes a value $x = A^*(r)$. (We handle the general case where the reduction can also choose the randomness below).

Thus, A^* implicitly defines a fixed distribution X such that calling A^* amounts to sampling an element x from X .

Since B only has black-box access to A^* , we can replace A^* with an oracle access to samples from X . Consider now the interaction between B^{A^*} and the challenger Γ , and let $x = (x_1, \dots, x_q)$ be the independent random samples from X received by B on calls to A^* . Note that q is at most t , the size of B .

For any fixing of x to a sequence $\bar{x} = (\bar{x}_1, \dots, \bar{x}_q)$ let $B^{\bar{x}}$ denote the execution of B where B is given the value $\bar{x}_i$ on the i^{th} call to the oracle. Note that $B^{\bar{x}}$ is of size at most $2t$, since it is hardcoded with the value $\bar{x}$.

Since X is a fixed distribution, by Equation 5.1 and an averaging argument there exists a fixed value $\bar{x} = (\bar{x}_1, \dots, \bar{x}_q)$ for which it holds that

$$\mathbf{Adv}_{\mathcal{G}}^{B^{\bar{x}}} = \Pr \left[\left(B^{\bar{x}} \leftrightarrow \Gamma \right) = 1 \right] - \gamma \geq \mu. \quad (5.2)$$

Which means that $\mathcal{G}$ is not $(2t, \mu)$ -secure.

We now deal with the more general case where the reduction B can choose the randomness given to A^* when making an oracle call. Let ℓ denote the number of random bits used by A^* , and let $\pi : \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ be a random permutation. Define $\hat{A}(r) := A^*(\pi(r))$ and note that $\mathcal{P}(\hat{A}) = 1$ iff $\mathcal{P}(A^*) = 1$ since A^* and $\hat{A}$ implicitly define the same distribution. Thus, by Equation 5.1 it follows that $\mathbf{Adv}_{\mathcal{G}}^{B^{\hat{A}}} \geq \mu$. Let $r_1, \dots, r_q \in (\{0, 1\}^\ell)^q$ be the random coins used by B when making calls to $\hat{A}$ in its interaction with Γ , and let $x_1 \dots x_q$ denote the answers. That is, the i^{th} query to $\hat{A}$ returns $x_i = \hat{A}(r_i)$. Without loss of generality, we can assume that $r_i \neq r_j$ for every $i \neq j$ (as there is no point in computing the same value twice). Since π is a random permutation, it follows that for every $i \in [q]$: $r_1, \dots, r_i, \hat{A}(r_1) \dots, \hat{A}(r_i)$ is distributed identically to $r_1, \dots, r_i, A^*(r_1^*), \dots, A^*(r_i^*)$ where $r_1^* \dots r_i^*$ are sampled independently conditioned on $r_j^* \neq r_{j'}^*$ for every $j \neq j'$. In particular, $(r_1^*, \dots, r_i^*)$ are independent from $(r_1, \dots, r_i)$, which means that $x_i = \hat{A}(r_i)$ is sampled from a fixed

distribution that does not depend on $(r_1, \dots, r_i)$. Thus, by an averaging argument similar to the argument for the simple case considered before, there exists a fixing of x to a sequence $\bar{x}$ such that $\text{Adv}_{\mathcal{G}}^{B_{\bar{x}}} \geq \mu$, implying that $\mathcal{G}$ is not $(2t, \mu)$ secure and completing the proof. $\square$

We derive the following corollary for our seedless primitives. For simplicity, we say that a keyless MCRH/condenser Cond is information theoretically broken if there exists a brute-force adversary A that contradicts the security of Cond .

Corollary 5.5. *Let Cond be a keyless MCRH/condenser candidate for parameters that are information-theoretically broken, and let $\mathcal{G}$ be any cryptographic game. If $\mathcal{G}$ is $(2t, \mu)$ -secure then there is no (t, μ) -black-box reduction showing the security of Cond from the security of $\mathcal{G}$.*

6 Condensers from Adaptive One-Way Functions

In this section, we construct seedless condensers for distributed sources from adaptive one-way functions. We start by defining adaptive one-way functions in Section 6.1 and give our construction in Section 6.2. We then discuss candidate instantiations for the adaptive one-way functions needed in our construction in Section 6.3.

6.1 Adaptive One-Way Functions

Adaptive one-way functions were introduced by [PPV08] to capture a family of functions that remain “one-way” even given an inverter for other functions. In this work, we consider a slight variant where the inverter outputs a *uniform* pre-image (note that this is an equivalent notion for permutations). Our construction only require pairs of adaptive one-way functions, so we focus on this special case. We start formalize the specific inverter we use and our notion of adaptive one-way security as follows.

Definition 6.1 (Uniform Inverters). Let $\mathcal{F} = \{f_b: \{0, 1\}^k \rightarrow \{0, 1\}^m\}_{b \in \{0, 1\}}$ be a pair of functions. We say that a randomized oracle algorithm $\mathcal{O}: \{0, 1\} \times \{0, 1\} \times \{0, 1\}^m \rightarrow \{0, 1\}^k$ is a *uniform inverter* for $\mathcal{F}$ if for every $b, b' \in \{0, 1\}$, and $y \in \{0, 1\}^m$, it holds that

- $\mathcal{O}(b, b', y) = \perp$ if $b' = b$ or $|f_{b'}^{-1}(y)| = 0$, or
- $\mathcal{O}(b, b', y)$ outputs a uniform value $x \leftarrow f_{b'}^{-1}(y)$ otherwise.

Definition 6.2 (Adaptive One-Way Functions). We say that a pair of efficiently computable functions $\mathcal{F} = \{f_b: \{0, 1\}^k \rightarrow \{0, 1\}^m\}_{b \in \{0, 1\}}$ is (t, μ) -*adaptive one-way for uniform inverters* if for any uniform inverter $\mathcal{O}$ for $\mathcal{F}$, t -size algorithm A , and $b \in \{0, 1\}$, it holds that

$$\Pr \left[\begin{array}{l} x \leftarrow \{0, 1\}^k \\ y = f_b(x) \\ x' \leftarrow A^{\mathcal{O}(b, \cdot, \cdot)}(b, y) \end{array} : x' \in f_b^{-1}(y) \right] < \mu.$$

In our construction, we require an additional statistical property bounding the regularity of the functions, formalized as follows.

Definition 6.3 (Two-Sided Bounded Regularity). Let $f: \{0, 1\}^k \rightarrow \{0, 1\}^m$ be a function with $k \geq m$. We say that f is *two-sided r -bounded regular* if for every $y \in \{0, 1\}^m$ it holds that

$$|f^{-1}(y)| \in \left[\frac{1}{r} \cdot 2^{k-m}, r \cdot 2^{k-m} \right].$$

We note that any permutation satisfies the above for $r = 1$.

6.2 Construction

We now give our a construction of a seedless condenser for distributed sources assuming a pair of adaptive one-way functions for uniform inverters that are also two-sided bounded regular, formalized in the following theorem.

Theorem 6.4 (Condensers from Adaptive One-Way Functions). *Let $k, m, t, r \in \mathbb{N}$, $\mu \in [0, 1]$, and $\mathcal{F} = \{f_b: \{0, 1\}^k \rightarrow \{0, 1\}^m\}_{b \in \{0, 1\}}$ be a pair of two-sided r -bounded regular functions. Assuming $\mathcal{F}$ is $(t + m + 1, \mu)$ -adaptive one-way for uniform inverters, then there exists a $(t, k \rightarrow k')$ -seedless condenser for distributed sources with $k' = \log(1/\mu) - 2\log(r)$.*

Proof. Given the pair of functions $\mathcal{F} = \{f_0, f_1\}$ as in the theorem statement, we define the seedless condenser Cond as follows:

$$\text{Cond}(x_0, x_1) := f_0(x_0) \oplus f_1(x_1).$$

It is clear from construction that Cond is efficiently computable as f_0 and f_1 are. It remains to argue that $\text{Cond}(X)$ has k' bits of min-entropy for any (t, k) -distributed source X generated by a t -size algorithm A . Namely, it suffices to show for any target value $z \in \{0, 1\}^m$ that

$$\Pr \left[\begin{array}{l} b \leftarrow A \\ x_b \leftarrow \{0, 1\}^k \\ x_{1-b} \leftarrow A(x_b) \end{array} : z = \text{Cond}(x_0, x_1) \right] < 2^{-k'}.$$

Note that without loss of generality, we may assume that b is fixed to either 0 or 1 by fixing the randomness for A .

Suppose by way of contradiction there exists a string $z \in \{0, 1\}^m$ such that A succeeds with probability greater than $2^{-k'}$ in the experiment above. By symmetry, it suffices to consider the case where $b = 0$, so it holds by assumption that

$$\Pr \left[\begin{array}{l} x_0 \leftarrow \{0, 1\}^k \\ x_1 \leftarrow A(x_0) \end{array} : z = \text{Cond}(x_0, x_1) \right] \geq 2^{-k'}. \quad (6.1)$$

Given such an algorithm A and string z , we construct an algorithm B against the adaptive-one-way security of f_1 . Recall that B has oracle access to a uniform inverter $\mathcal{O}(1, \cdot, \cdot)$ and takes as input an output $y = f_1(x) \in \{0, 1\}^m$ for $x \leftarrow \{0, 1\}^k$. We define B as follows:

Algorithm 6.5. $B^{\mathcal{O}(1, \cdot, \cdot)}(y)$:

1. Sample $x_0^* \leftarrow \mathcal{O}(1, 0, y \oplus z)$ from the uniform inverter.
2. Output $x_1^* \leftarrow A(x_0^*)$.

Note that since A is a t -size adversary, it follows that B runs in time $t + m + 1$, so B can succeed at inverting f_1 with at most μ probability by assumption. Using the fact that $\mathcal{O}(1, 0, y \oplus z)$ outputs a uniform pre-image from $f_0^{-1}(y \oplus z)$, note that the success probability of B is given as follows:

$$\Pr \left[\begin{array}{l} x \leftarrow \{0, 1\}^k \\ y = f_1(x) \\ x_0^* \leftarrow f_0^{-1}(y \oplus z) \\ x_1^* \leftarrow A(x_0^*) \end{array} : x_1^* \in f_1^{-1}(y) \right]. \quad (6.2)$$

Furthermore, it holds by construction that if $f_1(x_1^*) = y$, then $f_0(x_0^*) \oplus f_1(x_1^*) = \text{Cond}(x_0^*, x_1^*) = z$. So B 's success probability is given by

$$\Pr[\text{Cond}(x_0^*, x_1^*) = z].$$

We now relate the success probability of B to our assumption on the min-entropy of our condenser made in Equation 6.1. The key difference between the two probabilities is that the input x_0 to A is uniformly distributed in Equation 6.1, whereas the input x_0^* to A in Equation 6.2 is only “close” to uniform. Specifically, we show in the following claim that the distribution of x_0^* has multiplicative distance at most r^2 from uniform.

Claim 6.6. *Let x_0^* be distributed as in Equation 6.2. For every $a_0 \in \{0, 1\}^k$, it holds that*

$$\Pr[x_0^* = a_0] \in \left[\frac{1}{r^2} \cdot 2^{-k}, r^2 \cdot 2^{-k} \right].$$

Proof. Recall that $x \leftarrow \{0, 1\}^k$, $y = f_1(x)$ and $x_0^* \leftarrow f_0^{-1}(y \oplus z)$. We start by analyzing the distribution of y . By the two-sided r -bounded regularity of f_1 , it holds for every string $b \in \{0, 1\}^m$ that

$$\Pr[y = b] = \Pr[f_1(x) = b] = \frac{|f_1^{-1}(b)|}{2^k} \in \left[\frac{1}{r} \cdot 2^{-m}, r \cdot 2^{-m} \right]. \quad (6.3)$$

Now, to analyze the distribution of x_0^* , we have for any fixed value $a_0 \in \{0, 1\}^k$ that the following holds:

$$\begin{aligned} \Pr[x_0^* = a_0] &= \Pr[f_0(x_0^*) = f_0(a_0)] \cdot \Pr[x_0^* = a_0 \mid f_0(x_0^*) = f_0(a_0)] \\ &= \Pr[y \oplus z = f_0(a_0)] \cdot \Pr[x_0^* = a_0 \mid y \oplus z = f_0(a_0)] \\ &= \Pr[y = f_0(a_0) \oplus z] \cdot \frac{1}{|f_0^{-1}(y \oplus z)|}. \end{aligned}$$

In the last line above, we bound the first term by Equation 6.3, so

$$\Pr[y = f_0(a_0) \oplus z] \in \left[\frac{1}{r} \cdot 2^{-m}, r \cdot 2^{-m} \right].$$

We bound the second term by the two-sided r bounded regularity of f_0 , so

$$|f_0^{-1}(y \oplus z)| \in \left[\frac{1}{r} \cdot 2^{k-m}, r \cdot 2^{k-m} \right].$$

Combining the above, it follows that

$$\Pr[x_0^* = a_0] \in \left[\frac{1}{r^2} \cdot 2^{-k}, r^2 \cdot 2^{-k} \right],$$

as required for the claim. $\square$

Given the above claim, we are ready to finish the proof of the theorem. Consider the probability space in Equation 6.1 for $b = 0$ where $x_0 \leftarrow \{0, 1\}^k$ and $x_1 \leftarrow A(x_0)$. By Claim 6.6, it holds that for every $a_0 \in \{0, 1\}^k$,

$$\Pr[x_0^* = a_0] \geq \frac{1}{r^2} \cdot \frac{1}{2^k} = \Pr[x_0 = a_0] \cdot \frac{1}{r^2}.$$

Since $x_1^* = A(x_0^*)$ and $x_1 = A(x_0)$ by construction, it follows that for every pair $(a_0, a_1) \in \{0, 1\}^k \times \{0, 1\}^k$

$$\Pr[(x_0^*, x_1^*) = (a_0, a_1)] \geq \Pr[(x_0, x_1) = (a_0, a_1)] \cdot \frac{1}{r^2}.$$

By applying Cond to both joint distributions, the success probability of B from Equation 6.2 is given by

$$\begin{aligned} \Pr[x_1^* \in f_1^{-1}(y)] &= \Pr[\text{Cond}(x_0^*, x_1^*) = z] \\ &\geq \Pr[\text{Cond}(x_0, x_1) = z] \cdot \frac{1}{r^2} \\ &\geq 2^{-k'} \cdot \frac{1}{r^2}, \end{aligned}$$

where the last line follows from Equation 6.1. This implies that $\mu > 2^{-k'} \cdot r^{-2}$, so $k' \geq \log(1/\mu) - 2 \log r$, as required for the theorem. $\square$

6.3 Candidate Adaptive One-Way Functions

The work of [PPV08] introduced the notion of adaptive one-way functions and proposed candidates from a variety of algebraic assumptions. At a very high level, their candidates can be described via the following framework. For any one-way function $f: \{0, 1\}^{k+1} \rightarrow \{0, 1\}^m$, define the pair of functions $\{f_b: \{0, 1\}^k \rightarrow \{0, 1\}^m\}_{b \in \{0, 1\}}$ such that $f_b(x) = f(b||x)$. In general, this will *not* result in a secure pair of adaptive one-way functions, but security could be conjectured for certain one-way functions based on discrete-log, factoring, or other concrete assumptions.

Adaptive one-way functions from hardness in different axes. In this work, we suggest a different approach based on the idea of hardness in different axes from [LPS17], which is also implicitly used in other works constructing non-malleable primitives (see, e.g., [KS17, BDK⁺19, BL18, KK19, DKP21]). The main idea is to consider two separate axes X and Y of hardness that can be tuned independently of each other. Then the pair of adaptive one-way functions $\{f_0, f_1\}$ should have the property that f_0 is secure on axis X and broken on axis Y , whereas f_1 is broken on axis X but secure on axis Y .

Concretely, we consider a candidate pair of functions based on the additional axis of quantum hardness following [KK19]. To formalize the assumption we need, we define what we mean to be “broken” by a class of algorithms. Specifically, our notion of adaptive one-way functions stipulates that a function f is broken—or *inverted*—if an algorithm can uniformly sample inverses, which we formalize below.

Definition 6.7. We say that a function $f: \{0, 1\}^k \rightarrow \{0, 1\}^m$ is *inverted* by an algorithm A if for any $y \in \{0, 1\}^m$, $A(y)$ outputs a uniform element from $f^{-1}(y)$ or $\perp$ if $|f^{-1}(y)| = 0$.

Assumption 6.8. Let $k, m, t \in \mathbb{N}$ and $\mu \in [0, 1]$. There exists values $t_C, t_Q \in \mathbb{N}$ and a pair of one-way functions $\{f_0, f_1: \{0, 1\}^k \rightarrow \{0, 1\}^m\}$ such that

- f_0 is $(t \cdot t_Q, \mu)$ -secure for quantum circuits but is *inverted* by a t_C -size classical circuit, and
- f_1 is *inverted* by a t_Q -size quantum circuit but is $(t \cdot t_C, \mu)$ -secure for classical circuits.

Next, we show that the above assumption suffices for construction a pair of adaptive one-way function as in Definition 6.2.

Theorem 6.9. *Given Assumption 6.8 holds for values $k, m, t \in \mathbb{N}$ and $\mu \in [0, 1]$, then there exists a pair of functions $\mathcal{F} = \{f_0, f_1: \{0, 1\}^k \rightarrow \{0, 1\}^m\}$ that is (t, μ) -adaptive one-way for uniform inverters.*

Proof. Let $\mathcal{O}$ be any uniform inverter for $\mathcal{F}$ and A a t -size algorithm. We need to show that for $b \in \{0, 1\}$, it holds that

$$\Pr \left[\begin{array}{l} x \leftarrow \{0, 1\}^k \\ y = f_b(x) \\ x' \leftarrow A^{\mathcal{O}(b, \cdot, \cdot)}(b, y) \end{array} : x' \in f_b^{-1}(y) \right] < \mu.$$

We consider the cases of $b = 0$ or $b = 1$ separately.

When $b = 0$, we simulate every oracle query to $\mathcal{O}(0, \cdot, \cdot)$ as follows:

- To simulate queries $\mathcal{O}(0, 1, y)$, we use the t_Q -size quantum circuit the inverts f_1 to sample a uniform pre-image from $f_1^{-1}(y)$, and to simulate $\mathcal{O}(0, 0, y)$, we simply output $\perp$.

It follows that the output of $A^{\mathcal{O}(0, \cdot, \cdot)}(0, y)$ can be simulated by a quantum circuit of size at most $t \cdot t_Q$, so A can succeed with probability at most μ by assumption.

Next, when $b = 1$, we simulate every oracle query to $\mathcal{O}(1, \cdot, \cdot)$ similarly:

- To simulate $\mathcal{O}(1, 0, y)$, we use the t_C -size classical circuit that inverts f_0 to sample a uniform pre-image from $f_0^{-1}(y)$, and to simulate $\mathcal{O}(1, 1, y)$, we output $\perp$.

The output of $A^{\mathcal{O}(1, \cdot, \cdot)}(1, y)$ can be simulated by a classical circuit of size at most $t \cdot t_C$ in this case, so A can succeed with probability at most μ by assumption.

Thus, in either case, A succeeds with probability at most μ for any such uniform inverter oracle $\mathcal{O}$, so $\mathcal{F}$ is a (t, μ) -adaptive one-way function for uniform inverters, as required. $\square$

Candidate instantiations for Assumption 6.8. For the function $f_1: \{0, 1\}^k \rightarrow \{0, 1\}^m$, we propose using a function based on discrete-log in a fixed group like an appropriate elliptic curve group. With minimal padding, the group can be embedded as a permutation over $\{0, 1\}^m$, and the input can be arbitrarily extended to any length $k \geq m$ by appending additional ignored input bits. In terms of security, it is known that discrete-log can be broken by quantum circuits of size $\text{poly}(n)$ by Shor's algorithm [Sho97a]. On the other hand, for any $t < 2^{m/2}$, it is conjectured that such groups behave like a generic group, where the best inversion attacks have advantage $t^2 \cdot 2^{-m}$ [Sho97b].

For the function f_0 and some $c \in \mathbb{N}$, we propose using the c -wise parallel repetition of any post-quantum secure cryptographic hash function. Specifically, let $h: \{0, 1\}^{k/c} \rightarrow \{0, 1\}^{m/c}$ be a hash function for $k = m + 2c \cdot \log m$ and set

$$f_0(x_1, \dots, x_c) := h(x_1) \parallel \dots \parallel h(x_c).$$

We note that a random h will be two-sided m^2 -bounded regular with overwhelming probability, and if so, the repetition f_0 will be as well. We conjecture that the same will hold for a suitable cryptographic hash function h . In terms of security, f_0 can be inverted by a classical circuit in time $c \cdot m^2 \cdot 2^{m/c}$ by breaking the c -independent instances by brute-force. For any $t \ll 2^{m/c}$, we assume that each independent instance can be inverted with probability at most $t^2 \cdot 2^{-m/c}$ by a quantum circuit, matching best known generic quantum attacks [Gro96, CGLQ20]. We also assume that the c -wise parallel repetition of h admits no non-trivial attacks so that f_0 can be inverted with probability at most $t^2 \cdot 2^{-m}$.

Assuming essentially optimal security and statistical properties of the above two candidate functions, we get the following corollary to Theorems 6.9 and 6.4.

Corollary 6.10. *Let $c, m, k, t \in \mathbb{N}$ such that $4 \leq c < m$, $k = m + 2c \cdot \log m$, and $t < 2^{m/(2c)}$. Assuming the existence of a fixed group of size $[2^{m-1}, 2^m]$ where discrete-log is $(t, t^2 \cdot 2^{-m})$ -secure and the existence of a cryptographic hash function $h: \{0, 1\}^{k/c} \rightarrow \{0, 1\}^{m/c}$ that is two-sided m^2 -bounded regular and whose c -wise parallel repetition is $(t, t^2 \cdot 2^{-m})$ -post-quantum secure, then there exists a $(t, k \rightarrow k')$ -seedless condenser for distributed sources for $k' = m \cdot (1 - 2/c) - 2 \log t - O(\log m)$.*

Proof. Since f_0, f_1 are at most two-sided m^2 -bounded regular, then the corollary follows from Theorem 6.4 if $k' = \log(1/\mu) - 4 \log m \geq m \cdot (1 - 2/c) - 2 \log t - O(\log m)$, or $\mu \leq t^2 \cdot \text{poly}(m) \cdot 2^{-m+2m/c}$. We use Theorem 6.9 to analyze μ given our assumptions.

We have that $t_C = c \cdot m^2 \cdot 2^{m/c}$ and $t_Q = \text{poly}(m)$ in Assumption 6.8 based on the known attacks on our candidate functions. Then, it follows that f_0 is assumed to be $(t \cdot t_Q, t^2 \cdot \text{poly}(m) \cdot 2^{-m})$ -secure against quantum circuits, so $\mu \leq t^2 \cdot \text{poly}(m) \cdot 2^{-m}$. For our discrete-log-based function f_1 , it is assumed to be $(t \cdot t_C, t^2 \cdot c^2 \cdot m^4 \cdot 2^{2m/c} \cdot 2^{-m})$ -secure against classical circuits, so $\mu \leq t^2 \cdot \text{poly}(m) \cdot 2^{-m+2m/c}$. In either case, this suffices for the corollary. $\square$

7 Seedless Condensers from Correlation-Intractability

In this section, we show how to construct seedless condensers for distributed sources from one-way permutations and hash functions that satisfy a form of correlation-intractability (CI). We define the necessary primitives in Section 7.1. In Section 7.2, we construct a seedless condenser for distributed sources from CI hash functions and one-way permutations. Finally, in Section 7.3, we discuss existing candidates for the primitives used in our constructions.

7.1 Correlation-Intractable Hash Functions

CI is defined with respect to a class of relations, and, at a high level, CI stipulates that it is hard to find inputs with correlated outputs that satisfy some relation. We recall the definition of correlation-intractability below.

Definition 7.1 (Correlation-Intractable Hash Functions). Let $t \in \mathbb{N}$ and $\mu \in [0, 1]$. Let $R \subseteq \{0, 1\}^n \times \{0, 1\}^m$ be a relation. An efficiently computable function $h: \{0, 1\}^\lambda \times \{0, 1\}^n \rightarrow \{0, 1\}^m$ is (t, μ) -correlation-intractable for R if for every t -size algorithm A , it holds that

$$\Pr \left[\begin{array}{l} \text{hk} \leftarrow \{0, 1\}^\lambda \\ x \leftarrow A(\text{hk}) \\ y = h(\text{hk}, x) \end{array} : (x, y) \in R \right] < \mu.$$

For a set of relations $\mathcal{R}$, we say that h is (t, μ) -correlation intractable for $\mathcal{R}$ if it is μ -correlation intractable for every $R \in \mathcal{R}$.

We emphasize that in contrast to many of the seedless primitives we consider in this work, a CI hash function is a *keyed* or *seeded* primitive. Regardless, we show that it is still useful for constructing seedless condensers.

For simplicity, we define CI with respect to a uniformly random key, but we note that with minor modifications, we can instead rely on CI hash functions with structured hash keys.

We usually require that the CI-relation is sparse, which roughly says that for any input, most outputs should not satisfy the relation. Some level of sparsity is necessary to ensure that at least a random function satisfies CI for the relations. We formalize this requirement below.

Definition 7.2 (Sparse Relations). Let $\rho \in [0, 1]$. Let $R \subseteq \{0, 1\}^n \times \{0, 1\}^m$ be a relation. We say that R is ρ -sparse if for every $x \in \{0, 1\}^n$, it holds that

$$|\{y \in \{0, 1\}^m : (x, y) \in R\}| \leq \rho \cdot 2^m.$$

For our constructions, we only require CI for a restricted class of relations that can be efficiently recognized given both the input and output. We formalize property below.

Definition 7.3 (Recognizable Relations). Let $R \subseteq \{0, 1\}^n \times \{0, 1\}^m$ be a relation. We say that R is t -recognizable if there exists a t -size algorithm C such that

$$(x, y) \in R \Leftrightarrow C(x, y) = 1.$$

7.2 Construction

We now give our construction of a seedless condenser for distributed sources assuming a one-way permutation and a CI hash function for exponentially sparse and efficiently recognizable relations, formalized in the following theorem.

Theorem 7.4 (Condensers from CI Hash Functions). Let $k, m, t \in \mathbb{N}$ and $f: \{0, 1\}^m \rightarrow \{0, 1\}^m$, $h: \{0, 1\}^k \times \{0, 1\}^k \rightarrow \{0, 1\}^m$ be functions. Assuming f is a $(t + |h| + k, \mu)$ -secure one-way permutation and h is $(t + m, \mu)$ -correlation-intractable for 2^{-m} -sparse $(|f| + m)$ -recognizable relations, then there exists a $(t, k \rightarrow k')$ -seedless condenser for distributed sources for $k' = \log(1/\mu)$.

Proof. Let x'_0 denote the first m bits of x_0 . Given functions f, h as in the theorem statement, we define the seedless condenser Cond as follows:

$$\text{Cond}(x_0, x_1) := x'_0 \oplus f(h(x_1, x_0)).$$

It is clear from construction that **Cond** is efficiently computable as f and h are. It remains to argue that the output of **Cond**(X) has k' bits of min-entropy for any (t, k) -distributed source X . Let A be a t -size algorithm that generates X . It suffices to show for any target value $z \in \{0, 1\}^m$ that

$$\Pr \left[\begin{array}{l} b \leftarrow A \\ x_b \leftarrow \{0, 1\}^k \\ x_{1-b} \leftarrow A(x_b) \end{array} : z = x'_0 \oplus f(h(x_1, x_0)) \right] < 2^{-k'}.$$

Note that without loss of generality, we may assume that b is fixed to either 0 or 1 by fixing the randomness for the first call to A .

Suppose by way of contradiction that there exists a string $z \in \{0, 1\}^m$ such that A succeeds with probability at least $2^{-k'}$ in the experiment above. Let E_0, E_1 denote the events that A succeeds for b corresponding to either 0 or 1, respectively. Then, it must be the case that either E_0 or E_1 occurs with at least $2^{-k'}$ probability depending on which case holds. We consider each such case separately in the following two claims.

Claim 7.5. *Assuming f is $(t + |h| + k, \mu)$ -secure one-way permutation, then if $\Pr[E_0] \geq 2^{-k'}$, it holds that $k' \geq \log(1/\mu)$.*

Proof. Given A and the fixed string z , we construct an attacker B that takes as input $y \leftarrow \{0, 1\}^m$ and breaks that one-wayness of f as follows:

Algorithm 7.6. $B(y)$:

1. Set $x'_0 = z \oplus y$.
2. Sample $r \leftarrow \{0, 1\}^{k-m}$ and set $x_0 := x'_0 || r$.
3. Compute $x_1 \leftarrow A(x_0)$.
4. Output $x := h(x_1, x_0)$.

First, observe that B has size $t + |h| + k$ to run A and h and prepare the input x_0 . It follows that B can succeed with probability at most μ by assumption.

Next, note that whenever $z = \text{Cond}(x_0, x_1)$ it follows that $f(x) \oplus x'_0 = z$, so $f(x) = y$. Furthermore, $x'_0 = z \oplus y$ is uniformly distributed since y is, so x_0 is also uniformly distributed by construction. It follows that $z = \text{Cond}(x_0, x_1)$ and B succeeds with probability greater than $2^{-k'}$ by assumption. This implies that $2^{-k'} < \mu$, so $k' > \log(1/\mu)$, as required. $\square$

Claim 7.7. *Assuming h is $(t + m, \mu)$ -CI for 2^{-m} -sparse and $(|f| + m)$ -recognizable relations, then if $\Pr[E_1] \geq 2^{-k'}$, it holds that $k' \geq \log(1/\mu)$.*

Proof. Given A and the fixed string z , we construct an attacker B that breaks the correlation-intractability of h with respect to a relation R defined as follows:

$$(x, y) \in R \text{ if and only if } f(y) \oplus x' = z,$$

where x' is the first m bits of x . Note that R is 2^{-m} -sparse since f is a permutation and recognizable in size $(|f| + m)$ by construction. We define the attacker B that takes as input a hash key $x_1 \leftarrow \{0, 1\}^k$ and breaks that CI of h with respect to relation R by simply running A as follows:

Algorithm 7.8. $B(x_1)$:

1. Output $x_0 \leftarrow A(x_1)$.

It is clear that B has size $t + m$ since it simply runs A , so can succeed with probability at most μ by assumption.

Next, we observe that whenever $z = \text{Cond}(x_0, x_1)$, it holds that $f(y) \oplus x'_0 = z$ for $y = h(x_0, x_1)$. It therefore follows that $(x_0, y) \in R$, so B succeeds in breaking the CI of R with probability greater than $2^{-k'}$ by assumption. This implies that $2^{-k'} < \mu$, so $k' \geq \log(1/\mu)$, as required. $\square$

In either case from the above claims, it holds that $k' \geq \log(1/\mu)$, as required for the statement of Theorem 7.4. $\square$

7.3 Candidate CI Hash Functions from KDM-Secure Encryption

Our result from Theorem 7.4 relies on a correlation-intractable hash function with output length m for efficiently recognizable relations that are 2^{-m} -sparse—and hence each input has at most one output in the relation. There are two main candidates for such functions: one based on obfuscating pseudorandom functions [CCR16, KRR17], and one based on encryption schemes with key-dependent message (KDM) security [CCR18]. In this work, we focus on the hash functions from KDM security due to [CCR18] since this approach results in much simpler instantiations for concrete analysis.

Encryption schemes We first recall the syntax and correctness for a (secret-key) encryption scheme. An encryption scheme consists of an encryption algorithm Enc and a decryption algorithm Dec . For simplicity, we assume a message space of $\{0, 1\}^m$ and key and ciphertext space of $\{0, 1\}^k$ where $k \geq m$. An encryption scheme has the following syntax:

- $\text{ct} \leftarrow \text{Enc}(\text{sk}, \text{msg})$: The randomized encryption algorithm takes as input a secret key $\text{sk} \in \{0, 1\}^k$ and a message $\text{msg} \in \{0, 1\}^m$ and outputs a ciphertext $\text{ct} \in \{0, 1\}^c$.
- $\text{msg} \leftarrow \text{Dec}(\text{sk}, \text{ct})$: The deterministic decryption algorithm takes as input a secret key $\text{sk} \in \{0, 1\}^k$ and a ciphertext $\text{ct} \in \{0, 1\}^c$ and outputs a message $\text{msg} \in \{0, 1\}^m$.

Furthermore, (Enc, Dec) should satisfy the correctness property that for every $\text{sk} \in \{0, 1\}^k$ and $\text{msg} \in \{0, 1\}^m$, it holds that

$$\text{msg} = \text{Dec}(\text{sk}, \text{Enc}(\text{sk}, \text{msg})).$$

CI hash functions from KDM-secure encryption. Let (Enc, Dec) be an encryption scheme with key space and ciphertext space $\{0, 1\}^k$ and message space $\{0, 1\}^m$ as above. We consider the hash function

$$h(\text{hk}, x) := \text{Dec}(x, \text{hk}).$$

Namely, the hash function interprets the input x as a secret key and outputs the decryption of hk as if it were a ciphertext.

To prove that this construction satisfies CI for any sparse relation, [CCR18] requires that (Enc, Dec) satisfy key-dependent message (KDM) security and an additional “universal ciphertexts” statistical property. We formalize these two properties below.

Definition 7.9. An encryption scheme (Enc, Dec) is (t, μ) -KDM-secure if for all t -size algorithms A and any function $f: \{0, 1\}^k \rightarrow \{0, 1\}^m$, it holds that

$$\Pr \left[\text{sk} \leftarrow \{0, 1\}^k : A(\text{Enc}(\text{sk}, f(\text{sk}))) = \text{sk} \right] < \mu.$$

Definition 7.10. An encryption scheme (Enc, Dec) has η -universal ciphertexts the following two conditions hold:

- For every $\text{sk} \in \{0, 1\}^k$,

$$\Delta \left(\left\{ \begin{array}{l} \text{ct} \leftarrow \{0, 1\}^k \\ \text{msg} = \text{Dec}(\text{sk}, \text{ct}) \end{array} : \text{msg} \right\}, \{ \text{msg} \leftarrow \{0, 1\}^m : \text{msg} \} \right) \leq 2^{-(1+\eta) \cdot m}.$$

- For every $\text{sk} \in \{0, 1\}^k$, $\text{msg} \in \{0, 1\}^m$, it holds that

$$\Delta \left(\left\{ \begin{array}{l} \text{ct} \leftarrow \{0, 1\}^k \\ \text{s.t. } \text{Dec}(\text{sk}, \text{ct}) = \text{msg} \end{array} : \text{ct} \right\}, \{ \text{ct} \leftarrow \text{Enc}(\text{sk}, \text{msg}) : \text{ct} \} \right) \leq 2^{-(1+\eta) \cdot m}.$$

We note that [CCRR18] proves their construction secure w.r.t. relations that (1) may not be efficiently recognizable and (2) are only negligibly sparse rather than 2^{-m} -sparse. Below, we leverage 2^{-m} sparsity to get a tighter security reduction via their security proof, which also helps simplify their analysis. We give the proof for completeness.

Theorem 7.11 (Adapted from [CCRR18]). *Let $t, k, m \in \mathbb{N}$ such that $k \geq m$, $\mu \in [0, 1]$, and $\eta > 0$. Let (Enc, Dec) be a $(t, \mu - 2^{-\eta \cdot m + 1})$ -KDM-secure encryption scheme with η -universal ciphertexts with message space $\{0, 1\}^m$ and key and ciphertext space $\{0, 1\}^k$. Then, $h(\text{hk}, x) := \text{Dec}(x, \text{hk})$ is (t, μ) -correlation-intractable for all 2^{-m} -sparse relations.*

Proof. Let R be any 2^{-m} -sparse relation. Since the message space corresponding to the output y is $\{0, 1\}^m$, this implies for every input x corresponding to a secret key, there is a unique output y such that $(x, y) \in R$. Let f be the (potentially inefficient) function that on input some x outputs the unique y such that $(x, y) \in R$.

Suppose by way of contradiction that there exists a t -size algorithm A that wins the CI security game for relation R with probability greater than μ . We show that the same algorithm A wins in

the KDM security game for function f via the following sequence of inequalities:

$$\begin{aligned}
& \Pr \left[\begin{array}{l} \text{sk} \leftarrow \{0, 1\}^k \\ \text{ct} \leftarrow \text{Enc}(\text{sk}, f(\text{sk})) \end{array} : \text{sk} = A(\text{ct}) \right] \\
& \geq \Pr \left[\begin{array}{l} \text{sk} \leftarrow \{0, 1\}^k \\ \text{ct} \leftarrow \{0, 1\}^k \\ \text{s.t. Dec}(\text{sk}, \text{ct}) = f(\text{sk}) \end{array} : \text{sk} = A(\text{ct}) \right] - 2^{-(1+\eta) \cdot m} \\
& = 2^m \cdot \Pr \left[\begin{array}{l} \text{sk} \leftarrow \{0, 1\}^k \\ y \leftarrow \{0, 1\}^m \\ \text{ct} \leftarrow \{0, 1\}^k \\ \text{s.t. Dec}(\text{sk}, \text{ct}) = y \end{array} : \begin{array}{l} \text{sk} = A(\text{ct}) \\ \wedge f(\text{sk}) = y \end{array} \right] - 2^{-(1+\eta) \cdot m} \\
& \geq 2^m \cdot \left(\Pr \left[\begin{array}{l} \text{sk} \leftarrow \{0, 1\}^k \\ \text{ct} \leftarrow \{0, 1\}^k \end{array} : \begin{array}{l} \text{sk} = A(\text{ct}) \\ \wedge f(\text{sk}) = \text{Dec}(\text{sk}, \text{ct}) \end{array} \right] - 2^{-(1+\eta) \cdot m} \right) - 2^{-(1+\eta) \cdot m} \\
& \geq \Pr \left[\begin{array}{l} \text{ct} \leftarrow \{0, 1\}^k \\ \text{sk} \leftarrow A(\text{ct}) \end{array} : f(\text{sk}) = \text{Dec}(\text{sk}, \text{ct}) \right] - 2^{-\eta \cdot m + 1} \\
& = \Pr \left[\begin{array}{l} \text{hk} \leftarrow \{0, 1\}^k \\ x \leftarrow A(\text{hk}) \end{array} : (x, h(\text{hk}, x)) \in R \right] - 2^{-\eta \cdot m + 1}
\end{aligned}$$

We note that the second and fourth lines follow from the universal ciphertext properties of (Enc, Dec) , and the last line is simply a renaming of variables to match the CI security game. It follows that A wins the KDM security game with probability greater than $\mu - 2^{-\eta \cdot m + 1}$, in contradiction with our assumption since A is of size t . $\square$

Candidate KDM-secure encryption schemes. The work of [CCRR18] gives two candidates for KDM-secure encryption schemes with the universal ciphertexts properties. The first can be viewed as a bit-encryption variant of the ElGamal encryption scheme based on discrete-log. The second is a variant of Regev's encryption scheme based on LWE [Reg05]. For our eventual condensers, we require that either of these schemes have near-optimal KDM security for fixed choice of public parameters. We refer to Section 6 of [CCRR18] for full details on the exact assumptions.

We recall that our construction of Theorem 7.4 also relies on a one-way permutation, which we could instantiate with the discrete-log based candidate from Section 6.3. Based on these candidates, we get the following immediate corollary to Theorems 7.4 and 7.11.

Corollary 7.12. *Let $m, t \in \mathbb{N}$, $k = \text{poly}(m)$. Assuming the discrete-log based permutation over $\{0, 1\}^m$ is a $(t, \text{poly}(t) \cdot 2^{-m})$ -secure one-way permutation and either of the candidate encryption schemes from [CCRR18] with key and ciphertext space $\{0, 1\}^k$ and message space $\{0, 1\}^m$ are $(t, \text{poly}(t) \cdot 2^{-m})$ -KDM secure, then there exists a $(t, k \rightarrow k')$ -seedless condenser for distributed sources for $k' = m - O(\log t)$.*

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